Configure IPv6 addressing

Q-1 Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Ans: Practical: Assign and Verify a Static IPv6 Address

Objective: To assign a static IPv6 address to a network adapter and verify the configuration using the command line.

I'll use Windows as the example. Run these commands on your own computer.

Step 1: Open Command Prompt as Administrator

Press Windows + S, search for cmd, and select Run as administrator.

Step 2: Find Your Network Adapter

Run:

netsh interface ipv6 show interfaces

Note the name or interface index of your active network adapter, such as Wi-Fi or Ethernet.

Step 3: Assign a Static IPv6 Address

For a lab example, use the documentation-only IPv6 prefix 2001:db8:1::/64. Replace Wi-Fi with your actual adapter name.

netsh interface ipv6 add address interface="Wi-Fi" address=2001:db8:1::10

This adds the IPv6 address 2001:db8:1::10 to the selected adapter.

Note: This is an example address for practice, not a working public internet address. For real network connectivity, use an IPv6 prefix assigned by your network administrator or ISP.

Step 4: Verify the Configuration

Run:

ipconfig

Find your Wi-Fi or Ethernet adapter and look for the IPv6 address.

You can also use:

netsh interface ipv6 show addresses

Step 5: Record the Result

Item	Example Result
Network Adapter	Wi-Fi
IPv6 Address Added	2001:db8:1::10
Prefix Length	/64
Verification Command	ipconfig
Status	Confirm by checking your computer's output

Q-2 Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

Hint: Use the ping6 or ping -6 command for IPv6 testing.

Ans: Practical: Configure a Unique Local IPv6 Address and Test Connectivity

Objective: Configure a Unique Local IPv6 (ULA) address beginning with fd and test the connection to another device on the same LAN.

Step 1: Check Your Network Adapter

Open Command Prompt as Administrator in Windows and run:

netsh interface ipv6 show interfaces

Find your active adapter name, such as Wi-Fi or Ethernet.

Step 2: Assign a Unique Local IPv6 Address

For this example, use fd12:3456:789a:1::10. Replace Wi-Fi with your actual adapter name.

netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10

This adds the example IPv6 address to the adapter.

Step 3: Configure Another Device

The second device must also have a ULA address in the same subnet, for example:

- Device 1: fd12:3456:789a:1::10/64
- Device 2: fd12:3456:789a:1::20/64

Configure the second device with its own address. Do not assign the same address to both devices.

Step 4: Verify the IPv6 Configuration

Run:

ipconfig

Check that the selected adapter displays its configured IPv6 address.

Step 5: Test Connectivity

From Device 1, run:

ping -6 fd12:3456:789a:1::20

If the devices can communicate, you should receive replies from the second device. If the ping fails, check that both devices are on the same LAN, the addresses and prefix lengths are correct, and the firewall allows ICMPv6 traffic.

Result Table

Item	**Example**
Device 1 IPv6 Address	fd12:3456:789a:1::10/64
Device 2 IPv6 Address	fd12:3456:789a:1::20/64
Address Type	Unique Local IPv6 (ULA)
Testing Command	ping -6 fd12:3456:789a:1::20
Result	Record whether replies were received

Q-3 Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Constraint: Document the steps and the observed address behavior in a text file.

Ans: Practical: Set Up a Temporary IPv6 Address

Objective: Configure a temporary IPv6 address, check whether it changes after a reboot or network reconnect, and record the observations in a text file.

A temporary IPv6 address is commonly used for outgoing connections to improve privacy. On Windows, temporary IPv6 addresses are usually generated automatically when the network supports IPv6 privacy extensions. The address may change over time, but a reboot or reconnect does not guarantee that it will change.

Step 1: Check Your Current IPv6 Address

Open Command Prompt and run:

ipconfig

Find your active Wi-Fi or Ethernet adapter and record its IPv6 addresses.

Step 2: Enable Temporary IPv6 Addressing

Open Command Prompt as Administrator and run:

netsh interface ipv6 set privacy state=enabled

This enables IPv6 privacy addressing where supported.

Step 3: Check the Address Again

Disconnect and reconnect your Wi-Fi, or restart your computer. Then run:

ipconfig

Record the IPv6 address shown before and after. To inspect IPv6 addresses and their states in more detail, run:

Get-NetIPAddress -AddressFamily IPv6

Look for an address marked Temporary in the PrefixOrigin or related address information, where available.

Step 4: Document Your Observations

I created a text file with the steps and a section where you can record the actual results from your computer.

Q-4 Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Ans: Practical: Zomato-Style Restaurant Search Using IPv6

Objective: Create a small Python program that stores restaurant names and their IPv6 addresses. The user enters part of a restaurant name, and the program displays the matching restaurant and its IPv6 address.

Python Code

restaurants = [{"name": "Pizza Palace", "ipv6": "fd12:3456::1"}, {"name": "Burger House", "ipv6": "fd12:3456::2"}, {"name": "Spice Garden", "ipv6": "fd12:3456::3"}, {"name": "Royal Dosa", "ipv6": "fd12:3456::4"}]search = input("Enter restaurant name: ").lower()found = Falsefor restaurant in restaurants: if search in restaurant["name"].lower(): print("Restaurant:", restaurant["name"]) print("IPv6 Address:", restaurant["ipv6"]) found = Trueif not found: print("Restaurant not found.")

Example Output

Enter restaurant name: pizza

Restaurant: Pizza Palace

IPv6 Address: fd12:3456::1

How It Works

1. The program stores restaurant names and IPv6 addresses in an array (a list of dictionaries).
2. The user enters a partial restaurant name, such as pizza.
3. The lower() function makes the search case-insensitive.
4. The for loop checks each restaurant name for the entered text.

5. If a match is found, the program prints its name and IPv6 address. Otherwise, it displays Restaurant not found.

Q-5 Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Ans: ## Practical: IPv6 Subnetting and System Configuration

I have created a text file containing an AI-generated IPv6 subnetting scenario with 3 separate subnets, sample IPv6 addresses, Windows configuration commands, and a section to record your actual results.

Scenario Summary

| Department | IPv6 Subnet | Example Host Address |

| --- | --- | --- |

| Computer Lab | `2001:db8:1234:1::/64` | `2001:db8:1234:1::10/64` |

| Library | `2001:db8:1234:2::/64` | `2001:db8:1234:2::10/64` |

| Administration Office | `2001:db8:1234:3::/64` | `2001:db8:1234:3::10/64` |

Download the Text File

IPv6 Subnetting Scenario and Steps

Text file (.txt) — scenario, commands, and result template

Important: The `2001:db8::/32` range is reserved for documentation and practice. The file includes commands for adding an example address on Windows, but you must run them on your own system and record the actual output. For a working network connection, use an IPv6 prefix assigned to your network.